D07 · Warehouse Operations & WMS

04 · Labor management & workforce planning in the warehouse

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Warehouse labor management plans staffing from forecast volume and engineered standards, measures productivity against those standards, and develops the team through feedback, incentives and training.

2 WHY IT MATTERS

Labor is the largest controllable cost in a manual warehouse. **Without standards, there is no way to tell a slow day from a slow process, or to staff tomorrow's volume instead of yesterday's.** Turnover matters as much as productivity: new operators work below standard for months, so a team with high turnover is permanently on the learning curve.

3 HOW IT IS DONE

1. **Set engineered standards per task.** Time studies or WMS data for receiving, put-away, picking and packing.
2. **Staff to forecast volume.** Convert the volume plan into hours by shift, with a realistic utilization factor.
3. **Give individual, daily feedback.** Each operator sees performance against standard, not just the site average.
4. **Invest in retention.** Incentives, training and working conditions that keep experienced people.

IN PRACTICE

Implement individualized productivity feedback — not just the team average

An operator who sees their daily productivity vs. standard has information to improve. A team that only sees the DC average has no individual reference to act on.

Volume-based staffing model is better than fixed staffing

A DC that adjusts headcount to projected daily volume (more operators on high-volume days, fewer on low-volume days) has a structurally lower labor cost than one with fixed staffing for the worst case.

The productivity incentive program has the best ROI of any warehouse management initiative

A well-designed incentive program (bonus for LPPH above standard, top performer recognition) improves average team productivity 10–20% with an incentive cost typically <3% of labor cost.

Talent retention in the warehouse is as important as productivity

Warehouse turnover rate impacts productivity because new operators are at 60–80% of standard during the first 3 months. A warehouse with 40% annual turnover always has 20–25% of its team on a learning curve.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Labor Management System implementation — DC with 120 picking operators — The DC has no individual productivity visibility. LPPH variability between operators is enormous.

Pre-LMS metrics	Value	Post-LMS action and result
LPPH range between operators (best vs. worst)	HIGH	42–98 LPPH (133% variation) · No individual feedback
High-productivity operators identified and recognized	HIGH	0% — all treated equally · No incentive system

Pre-LMS metrics	Value	Post-LMS action and result
Productive vs. presence time (utilization)	MEDIUM	68% average utilization (32% unmanaged idle time)
Productivity-based incentive program	HIGH	No program · High turnover: 38%/year

Result: Post-LMS (12 months): Average LPPH: 64 -> 82 (+28%). Utilization: 68% -> 79%. Turnover: 38% -> 24%. Daily individualized feedback and the incentive program (weekly bonus for operators above standard) generated the greatest impact. Operators with consistently low LPPH received targeted training.

HOW IT IS MEASURED

Labor Utilization % (% of operator presence time that is productive)

Labor Utilization %(Productive operator hours / Total paid presence hours) × 100

Benchmark: 75–85% utilization in well-managed manual picking operations Utilization <70% indicates significant idle time (waits, unassigned operators, flow problems) being paid without generating value.

LPPH vs. Engineered Standard % (actual productivity vs. calculated standard)

LPPH vs. Engineered Standard %(Actual LPPH of operator or shift / Calculated standard LPPH) × 100

Benchmark: >95% of standard for experienced operators · 80–90% during the learning curve (first 3 months) Comparing actual LPPH vs. engineered standard identifies operators below standard who need additional training, and those above who deserve recognition.

WHAT YOU WOULD USE

Labor Management

Manhattan Associates WMS — Labor Management — Module: Engineered Standards + LMS + Incentive ManagementManhattan Associates labor management module with per-task productivity standards, real-time individual measurement, and incentive program management.

SAP EWM — Labor Management — Module: WMS-Integrated Labor TrackingSAP EWM labor management module integrated with the WMS. Measures productivity per operator, per task, and per shift.

4 SO WHAT

Plan, measure and keep your people. Engineered standards, volume-based staffing, daily individual feedback and a retention strategy are the cheapest productivity program a warehouse has.

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